Creating IBM Cloud Functions Entities

Estimated time needed: 45 minutes

In this lab you will begin creating IBM Cloud Functions entities! You will create and invoke actions and sequences, and you will be encouraged to explore the web action and API management capabilities provided by Cloud Functions. All this will familiarize you with how to start using IBM Cloud Functions yourself.

> Several labs/assignments in this course involve the usage of IBM Cloud account, which did not previously require credit card registration. However, as of October 1st, 2021, IBM Cloud has stopped providing Lite accounts. In the meanwhile, if you don't want to disclose your payment card information to create an IBM Cloud account, you can create an account with a feature code that offers you a 252-day trial account. You can obtain the IBM cloud feature code from the course in the Activate Trial Account section.

Also, please create an IBM cloud account in the US region.
However, if you decide to create an IBM Cloud account with a credit card, it is recommended that you specify US dollars for billing for faster approval. The labs/assignments in the course are designed to be completed with free tiers or Lite plans to avoid costs. Be aware that your credit card may be charged if you exceed the free usage, so it is a good practice to delete your instances once they are not required. And in case of any services that are billed on an hourly basis, please be sure to stop or delete your instance as soon as your lab is complete.
Note: the IBM and Coursera course teams are not responsible for any billed usage you may incur while using IBM Cloud.

Objectives

After completing this lab, you will be able to:

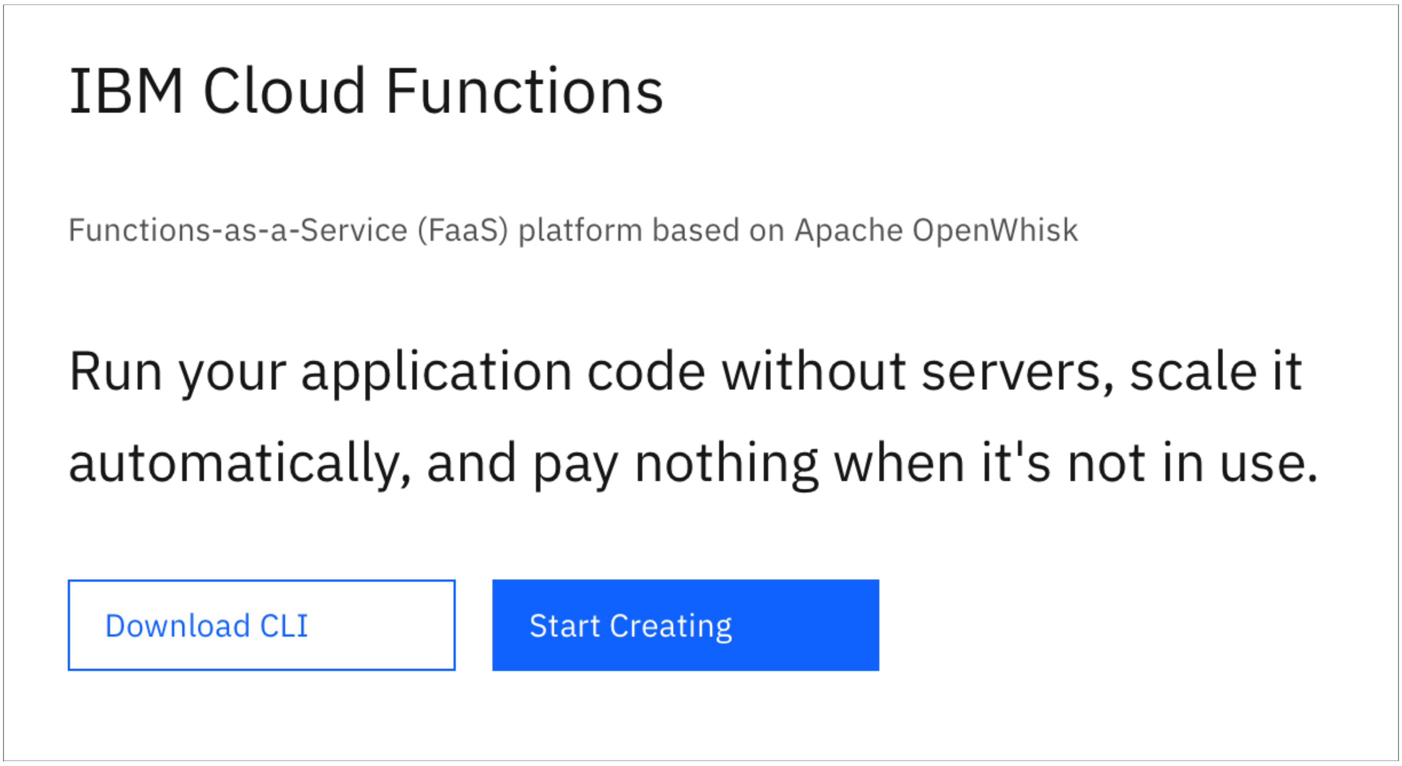
1. Create and invoke an action
2. Use custom Node.js code for an action
3. Create and invoke a sequence
4. Enable a web action and access an action using HTTP
5. Test API for your action/Sequence

Exercise 1 : Create and Invoke an Action

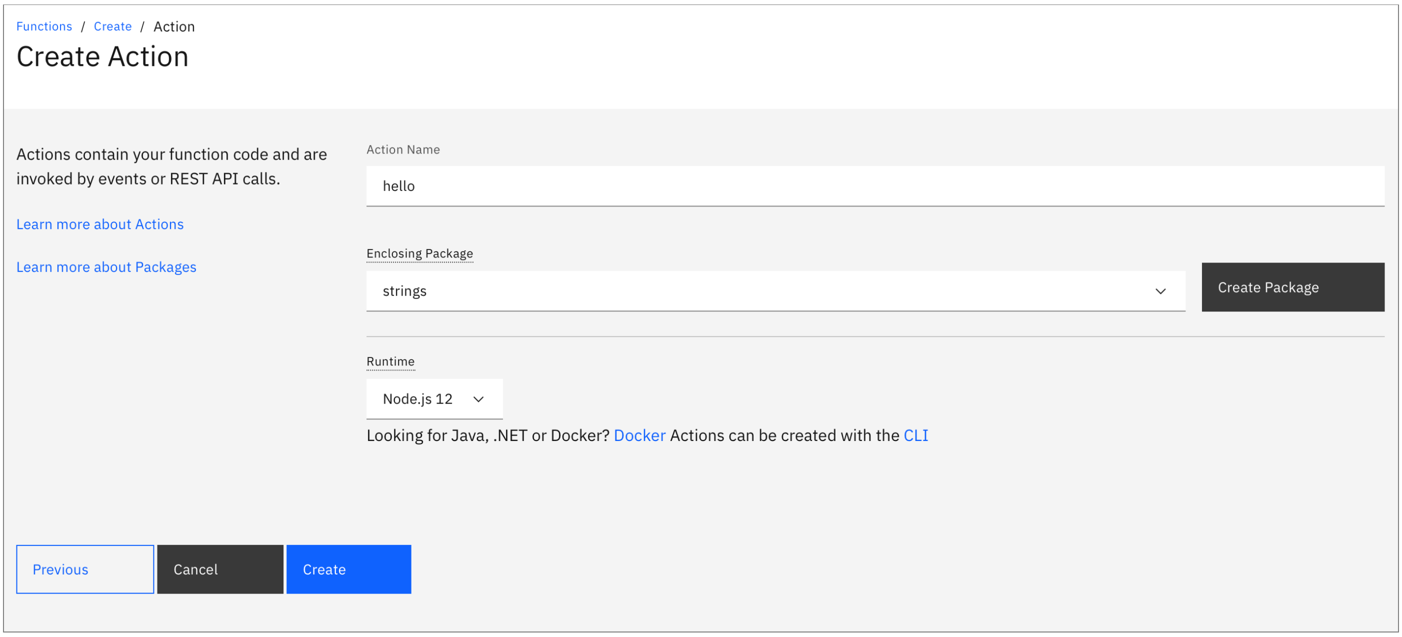
In this exercise, you will create and invoke your first IBM Cloud Functions action.

1. Go to the IBM Cloud Functions UI.

2. Click **Start Creating**.

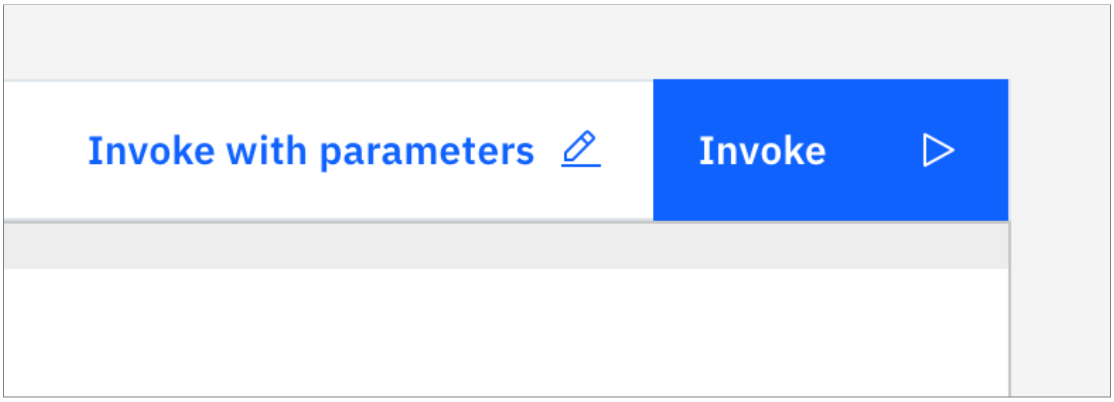


1. Click **Action** in order to create an action.
2. Name your action **hello**.
3. Click **Create Package** in order to put this action into a package.
4. Name the package **strings** since we'll make several actions that perform string manipulation.
5. Click **Create**.
6. Choose a Node.js runtime.
7. Click **Create**.



Now you have created an action! Your newly created action should be pre-populated with Node.js code to return a hello world message.

1. Click **Invoke**.

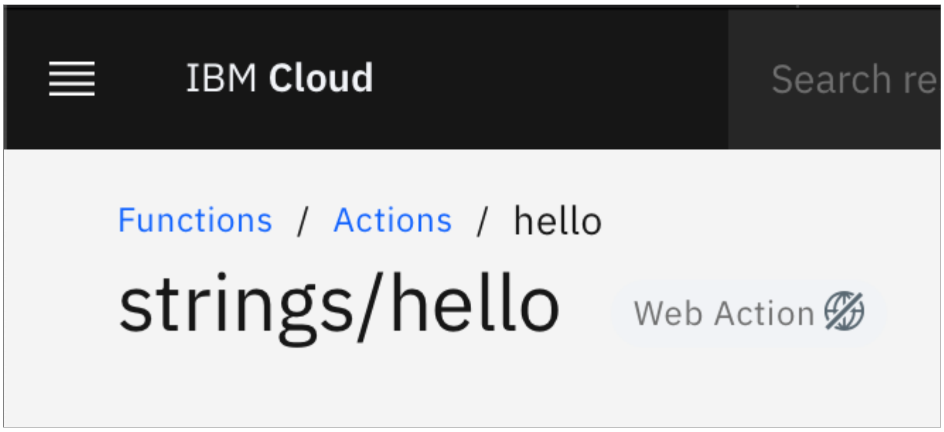


The activations should appear in a panel on the right. Wait for a second and the results should appear within your activation.

Exercise 2 : Create a Second Action

In this exercise you will create a second action that performs a slightly more advanced task.

1. Click **Actions** in the breadcrumbs on the top right to return to the actions page. When it loads, you should see your **strings** package with your **echo** action in it.



2. Click **Create** and then **Action** to create another action.
3. Name your action **capitalize**.
4. Select **strings** as the enclosing package.
5. Choose a Node.js runtime.
6. Click **Create**.
7. Use the following code for the main function in the code editor.

```
function main(params) {
  return { message: params.message.toUpperCase() };
}
```

8. Click **Save**.
9. Click **Invoke with parameters**.
10. Put the following in the input box:

```
{
  "message": "I'm hungry!"
}
```

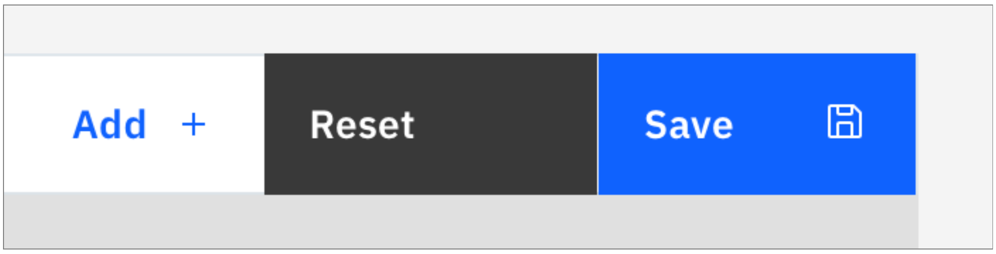
11. Click **Apply**.
12. Click **Invoke**.
13. Check the activation results to see your message capitalized.

Exercise 3 : Create and Invoke a Sequence

In this exercise you will create a sequence using your two previously created actions. You'll then invoke the sequence.

1. Click **Actions** in the breadcrumbs on the top right to return to the actions page. When it loads, you should see your **strings** package with your **hello** action in it.
2. Click **Create** and then **Sequence**.
3. Name the sequence **hello_capitalize**.
4. Select **strings** as the enclosing package.
5. Select an existing action for this sequence and choose **strings / hello**.
6. Click **Create**.
7. Click **Add +** to add another action to this sequence.

8. Click **Select Existing** and choose **strings / capitalize**.
9. Click **Add**.
10. Click **Save** to save the sequence.



11. Click **Invoke with parameters**.
12. Put the following in the input box:

```
{
  "text": "Testing my sequence."
}
```

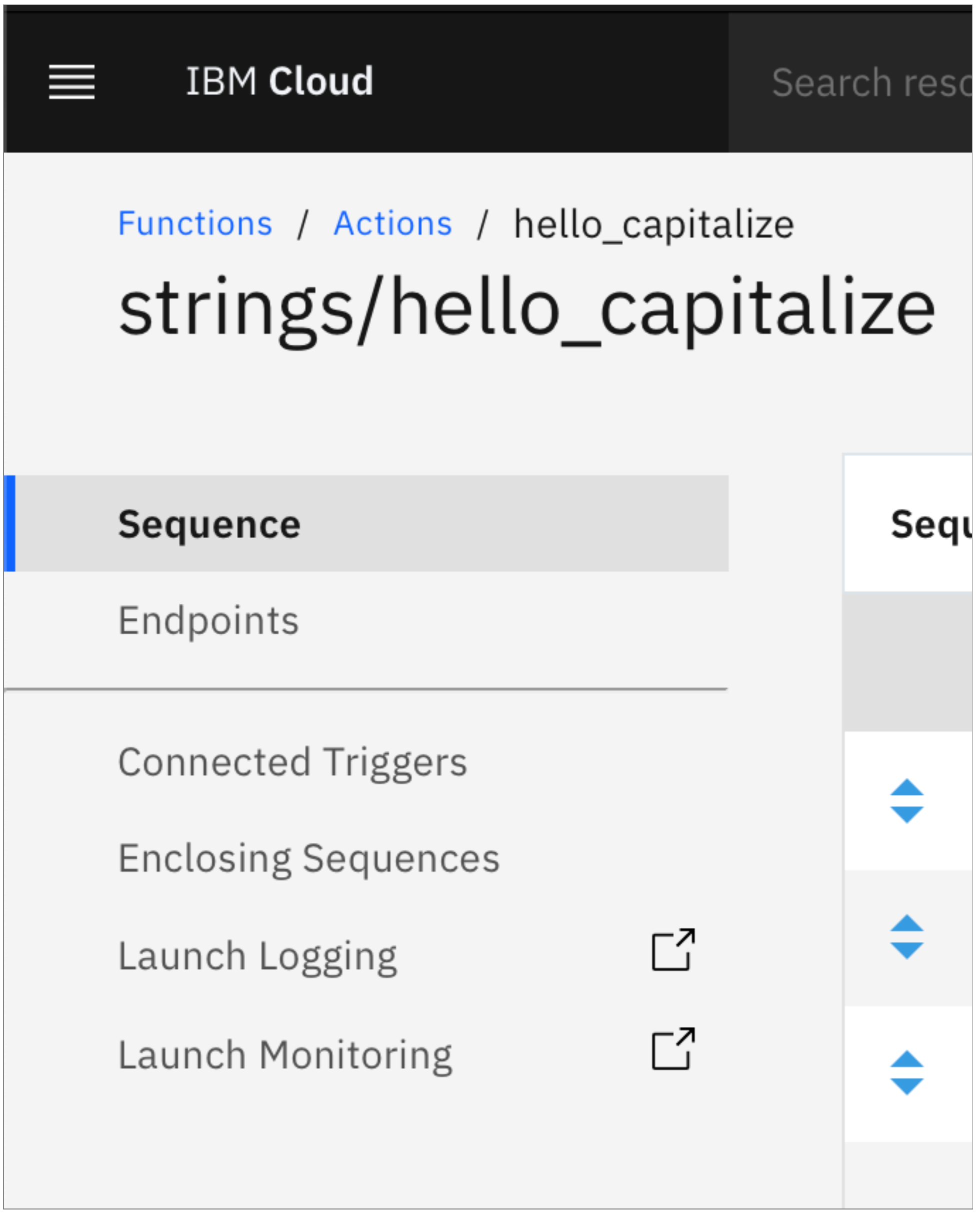
13. Click **Apply**.
14. Click **Invoke**.
15. Check the activation record and ensure that the results include this message: "TESTING MY SEQUENCE."

Great job! You passed the provided input to your sequence. This input was echoed, and that output was passed to the capitalize action, which capitalized the whole thing.

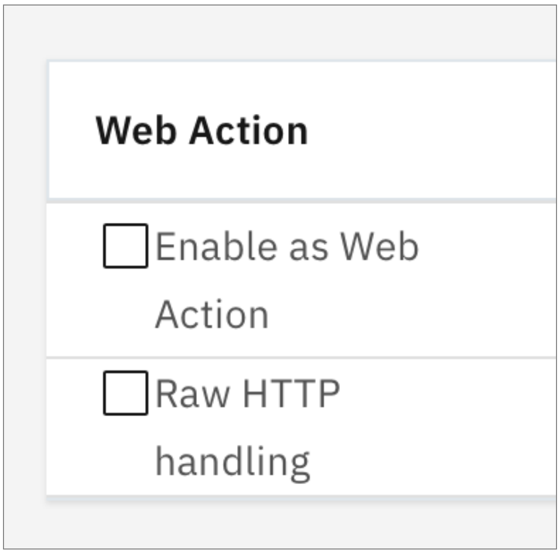
Exercise 4 : Enable Web Actions

In this exercise you will enable a web action so that your action can be called using HTTP.

1. From the **string/hello_capitalize** sequence page, click **Endpoints** in the navigation menu.



2. Click the checkbox for **Enable as Web Action**.



3. Click **Save**.

4. Copy the `curl` command located on this page.

5. Open the IBM Cloud Shell at the top right of the cloud console.

6. Run `ibmcloud iam oauth-tokens` to list your IAM tokens for authentication.

7. Copy the bearer token. You can ignore the word "bearer" and just copy the token.

8. Plug the token into the `curl` command that you copied.

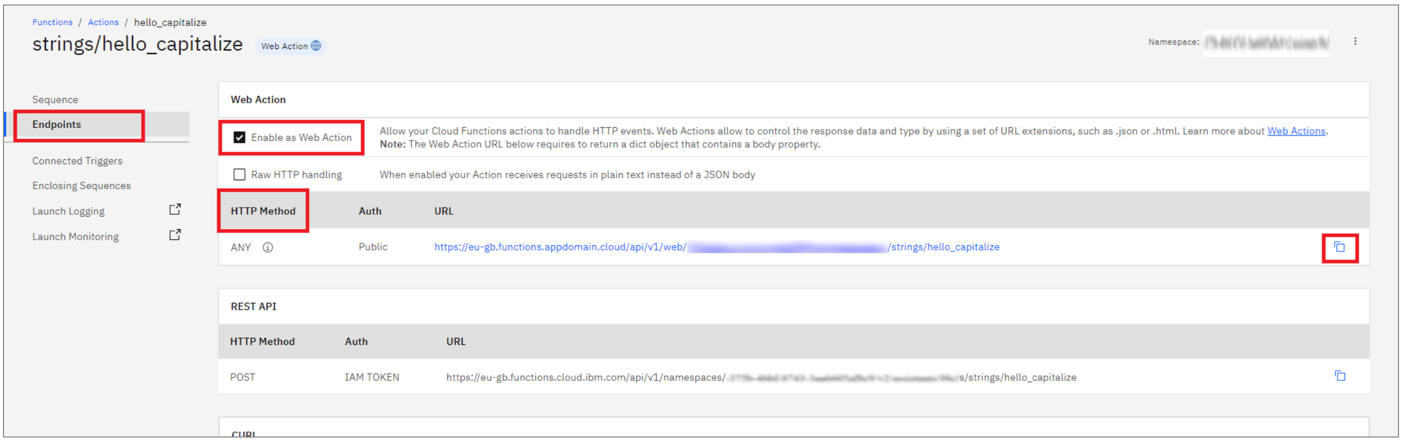
9. In the Cloud Shell, run the `curl` command with your bearer token plugged in.

Exercise 5 : API Management

In this exercise, you will test API to invoke your actions/sequence.

1. Return to the main [IBM Cloud Functions page](#).

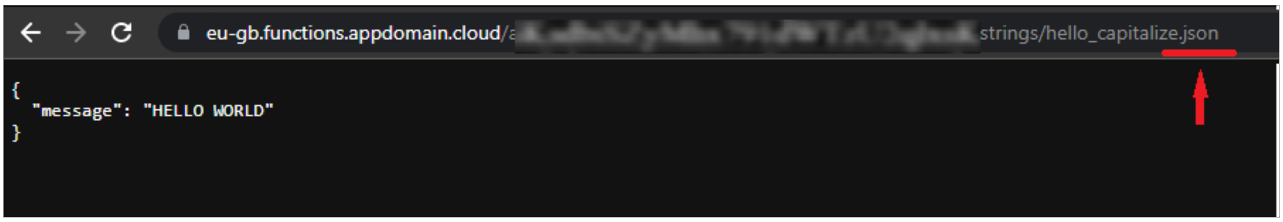
2. Click **Actions** and go to the sequence **hello_capitalize**. Then click on endpoints and copy the **HTTP Method** URL



> **Note:** Please note that using Public URL is not recommended for Production. In real time scenarios please use the URL with the IAM Authentication for security purpose.

3. Now, open a new browser tab and paste the copied URL into your browser and add **.json** at the end of the URL. And hit enter.

You should see your capitalized hello world message in your browser!



Conclusion

In this lab you created your first actions and sequence! You created two actions that can be independently invoked, and you also joined those actions into a sequence that you invoked. You then enabled a web action and invoked the action using HTTP in a terminal. Finally, you created an API to invoke your actions/sequence.

Author(s)

Alex Parker

Changelog

Date	Version	Changed by	Change Description
08-Sept-2022	1.1	Samaah Sarang	Instructions changed